Study Notes: Operating Systems

Section 1: Processes

A process is a program in execution. Process scheduling allocates CPU time among ready processes. Context switching saves and restores the execution state.

Section 2: Paging and Page Replacement

Paging divides virtual memory into fixed-size pages and physical memory into frames. A page table maps each virtual page to a physical frame.

A page fault occurs when a process accesses a page that is not currently in physical memory. The operating system loads the page from storage before execution continues.

Page replacement chooses a resident page to evict when memory has no free frame. FIFO replaces the page that arrived earliest. LRU replaces the page that has not been used for the longest time. The Clock algorithm approximates LRU using a reference bit and a circular hand.

Section 3: File Systems

A file system organizes files and directories on persistent storage.